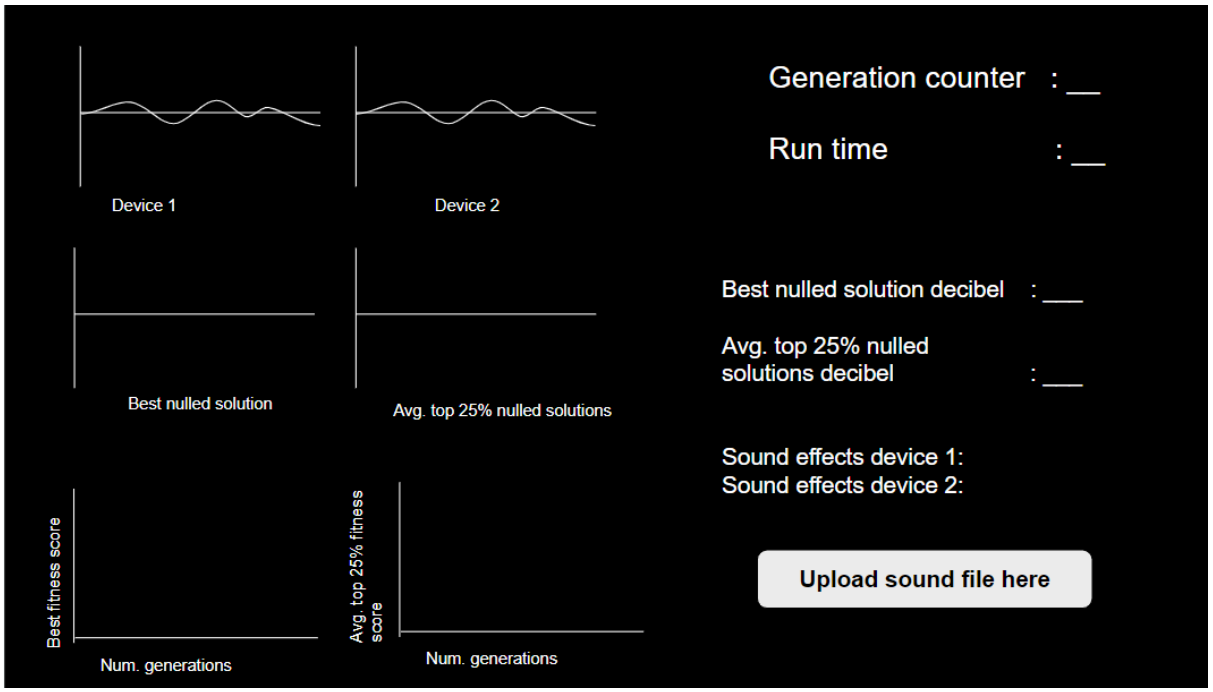
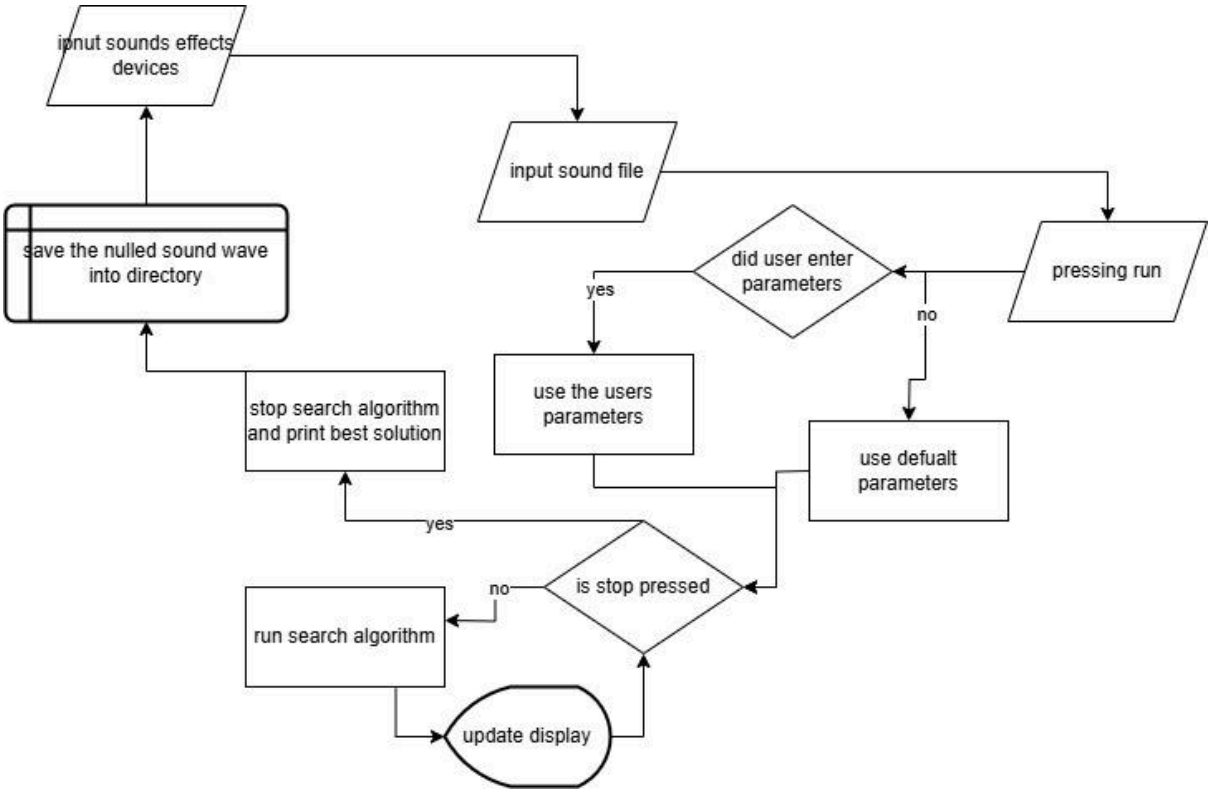
Criterion B: Design

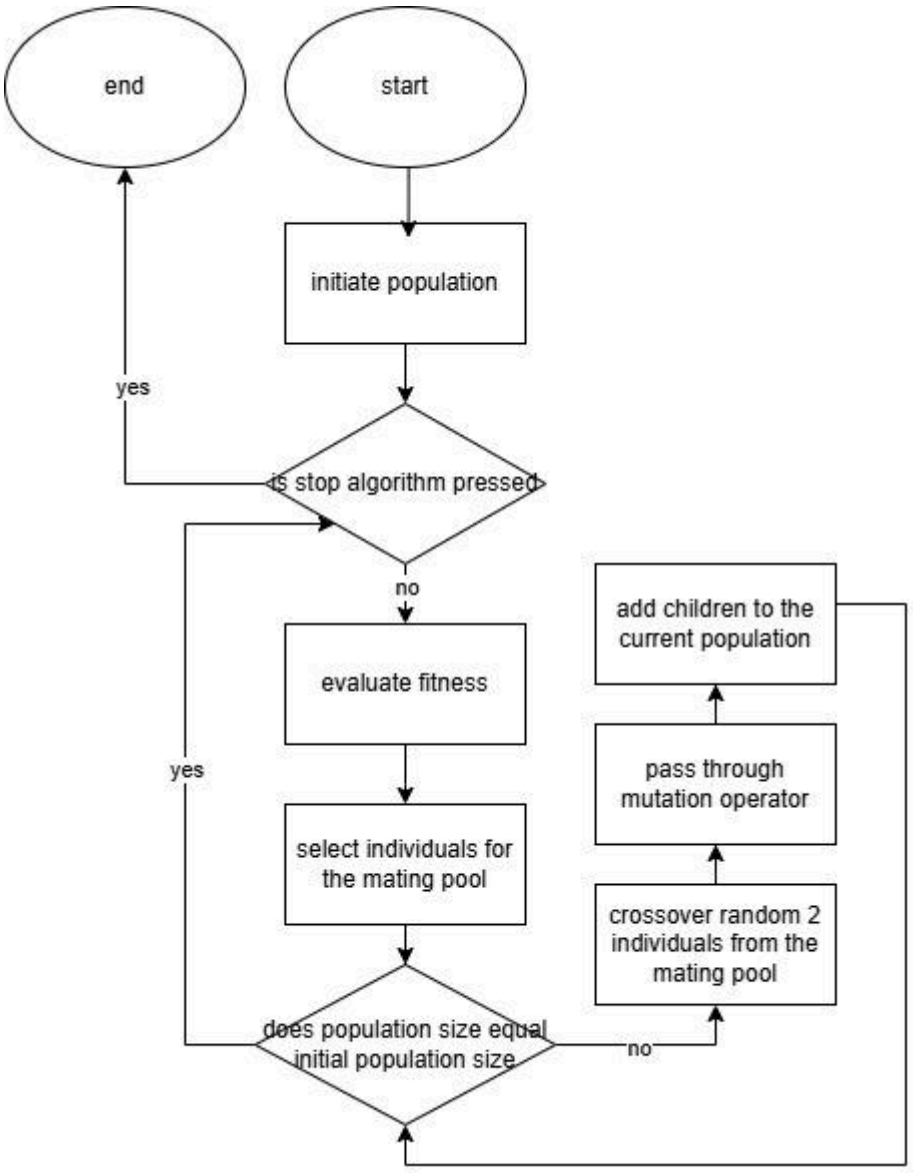
GUI initial design idea



System Flow Chart



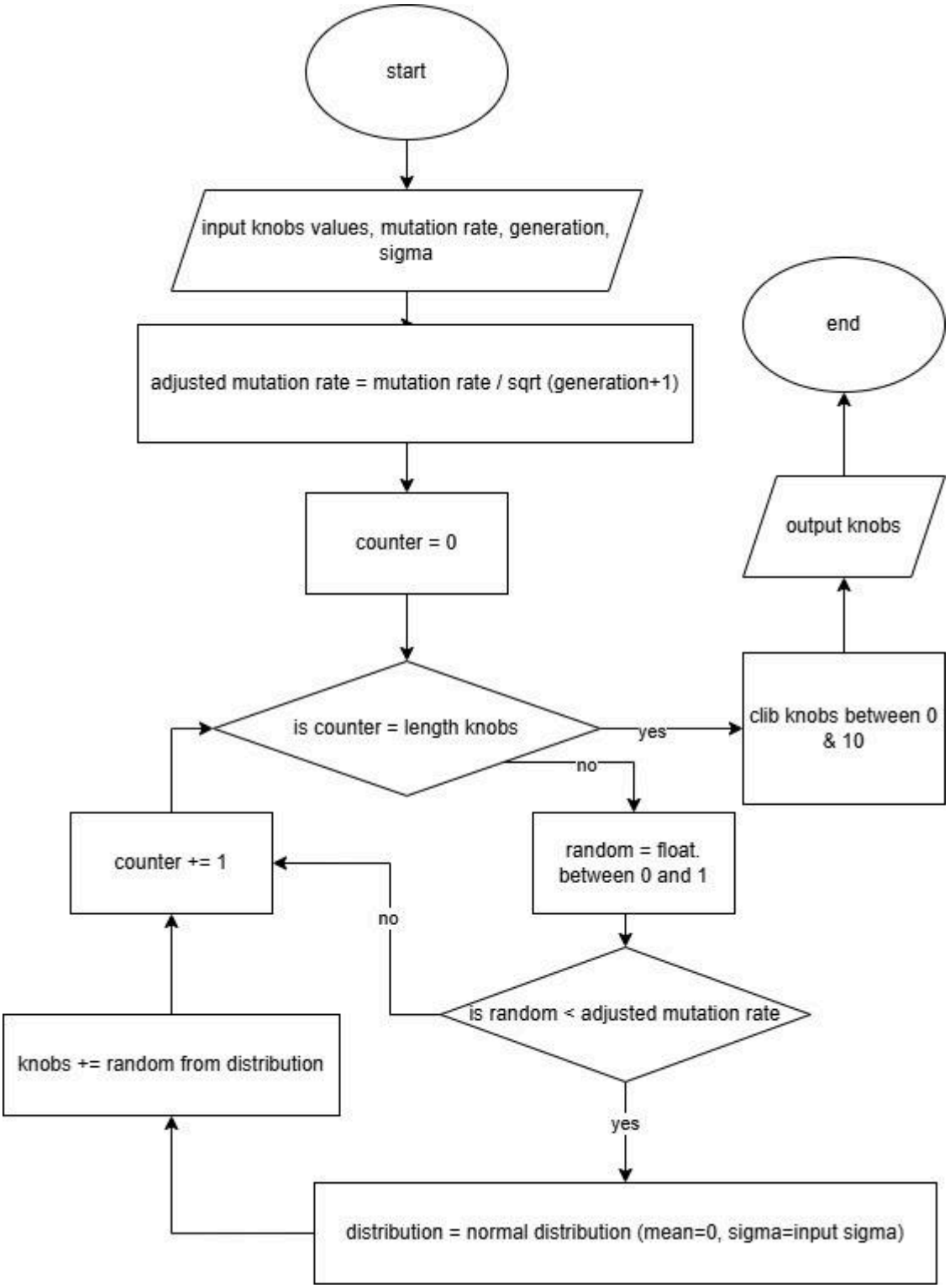
Genetic algorithm system flow chart



Main Functions

Function	**Category**	**Description**
UI initialization	UI	This sets up the display, the graphs, input text boxes, buttons… etc
Run Algorithm	system	Checks input parameters, initializes the population, and starts the timer and calls the “generation step” function.
Generation step	system/UI	This calls “run generation” function and retrieves data from the generation and plots it
Run generation	GA	This function calls loops through one generation, evaluating mating pool, selection, crossover, and mutation.
Fitness Evaluation	GA	Calculates the RMSE of 1 individual
Selection methods	GA	Selects individuals from the population, creating a mating pool
Crossover	GA	Selects two individuals from the mating pool and perform crossover, introducing new individuals
Mutation	GA	This mutates individuals, introducing some randomness into the algorithm to explore new solutions.

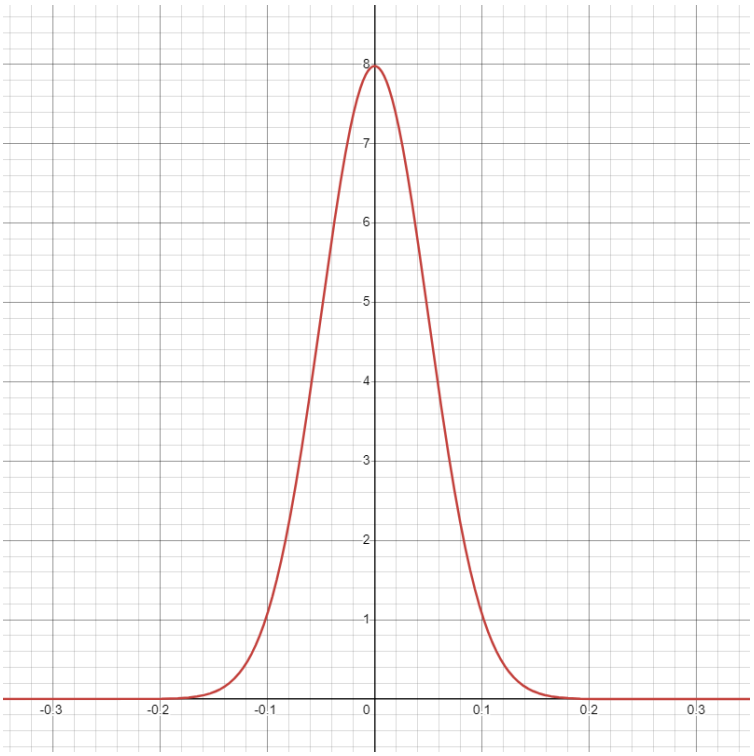
Mutation operation program flow chart



$$f(x) = \frac{1}{a\sqrt{(2\pi)}} e^{-0.5\left(\frac{(x-b)}{a}\right)^2}$$

a = sigma (standard deviation of normal distribution)

b = mean (mean of the normal distribution)

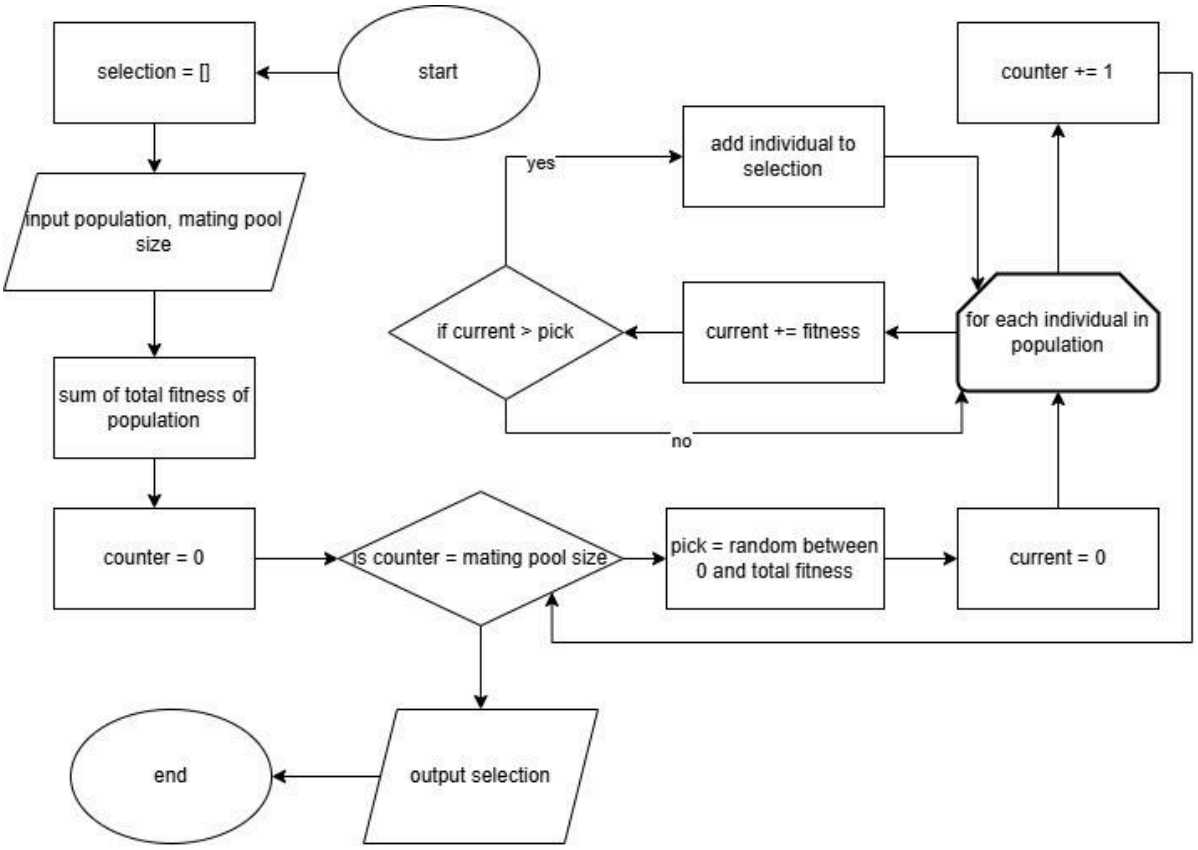


(Desmos graphing calculator)

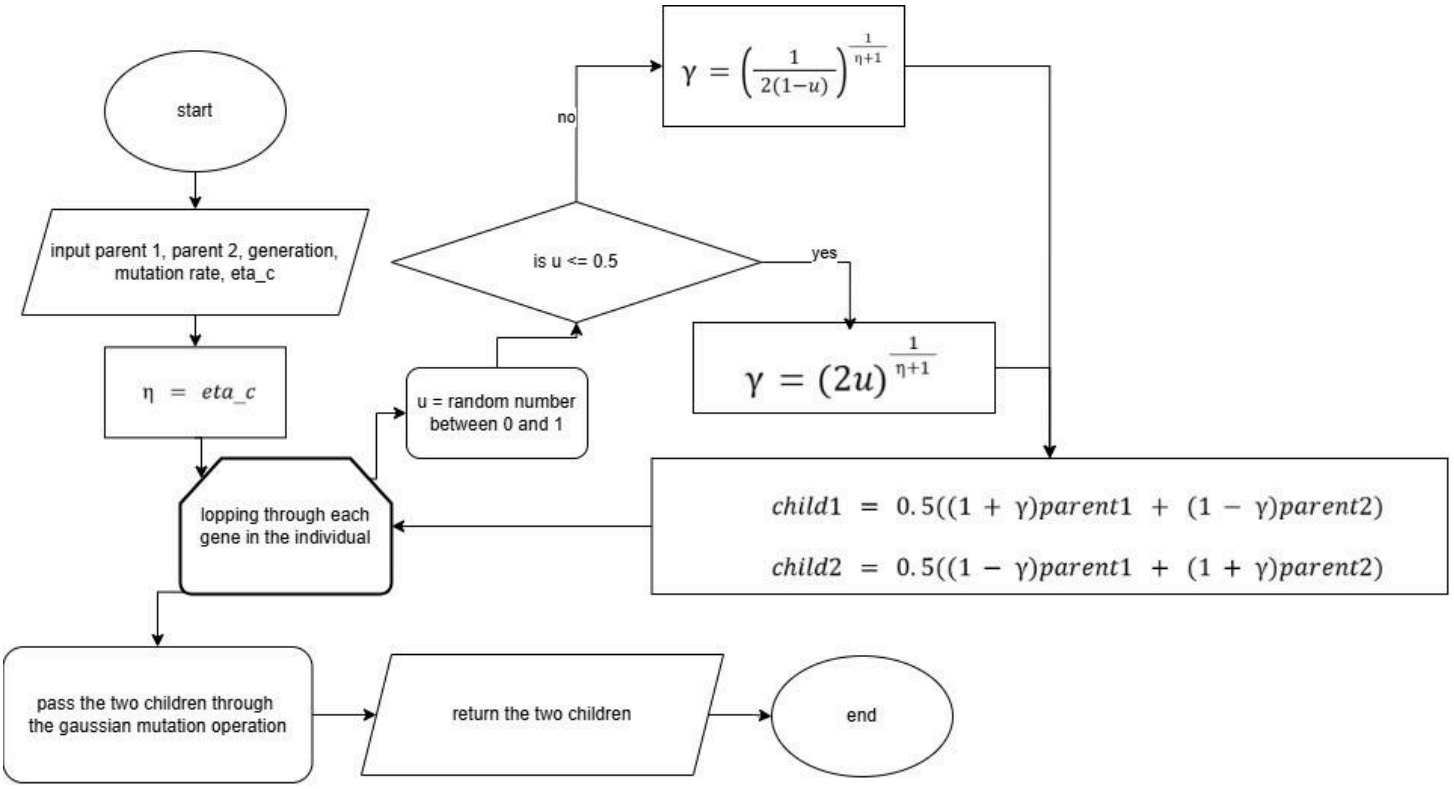
This means most values of mutation would be around 0, a very very low probability of values being larger than ±0.15.

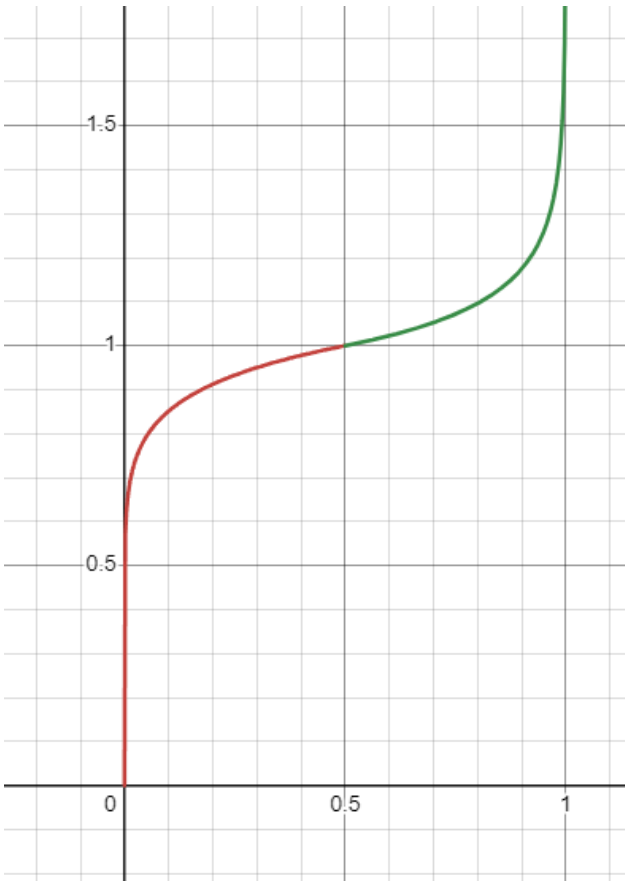
The value of sigma will need to be optimized

Roulette wheel selection method program flowchart



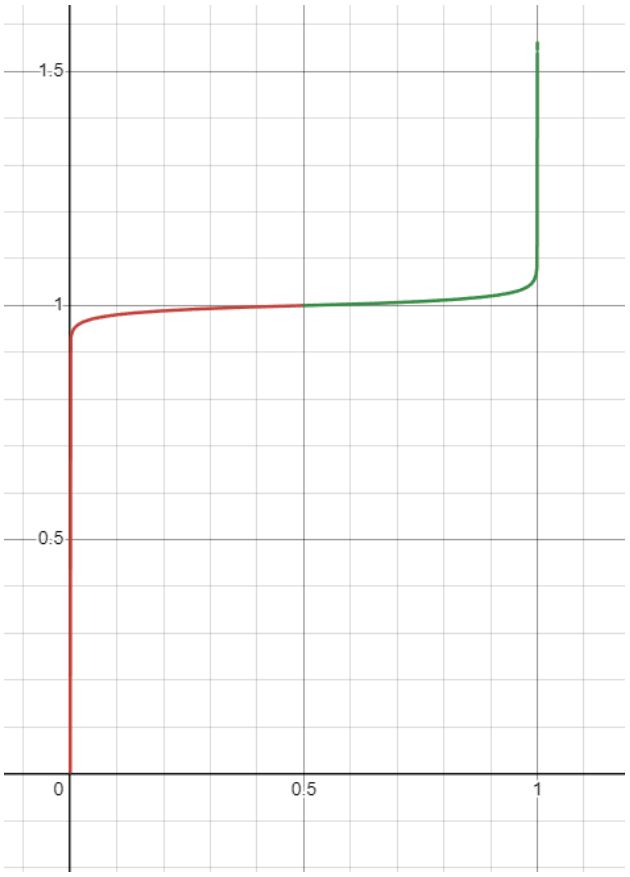
Simulated binary Crossover(SBX) program flowchart





Here eta_c = 9, a low value of eta_c produces values further away from 1 for betaq, meaning the values of the children will deviate further away from the parents.

(Desmos Graphing Calculator)



Here eta_c = 80, a larger value of eta_c gives values closer to 1 for betaq. Meaning the children will be closer to their parents.

The value of eta_c will need to be optimized

(Desmos Graphing Calculator)

Success Criteria	**Test type**	**Nature of test**	**Expected outcome**
Can choose any input sound	Dynamic testing	Allows the user to input the sound of their choice to be ran through the search algorithm	Desired input sound from user is displayed
A search algorithm attempting to to replicate one device to another	Dynamic testing	The algorithm is tweaking the knobs in attempt to minimize the difference between them	The fitness function should decrease, the nulled solution should slowly diminish as the algorithm proceeds
Selection method	Dynamic testing	Choosing the fitter individuals to pass on genes, however still allowing diversity	A mating pool is chosen to be able to reproduce and pass on genes
Crossover operation	Dynamic testing	Ensure successful creation of children	New individuals created, similarity of children to parent is based on the distribution index
Mutation operation	Dynamic testing	Ensures some individual's genes are mutated	Some genes have been changed, the difference in change is based on the standard deviation
The user is allowed to change the search parameters	Dynamic testing	Ensures the parameters entered are used in the Genetic algorithm	The chosen parameters by the users are used in the genetic algorithm
Efficient genetic algorithm (optimized parameters)	Dynamic testing	Measures performance of algorithm and to change parameters accordingly	Lowest fitness is achieved in the quickest time
GUI is simple and easily understandable	Beta testing / User acceptance testing	Ensures the GUI is easily understandable	The user is able to operate the search algorithm with ease and minimal confusion